

Mounica Paladugu

meismounica079@gmail.com | 206-591-5835 | Mountain View, CA
[linkedin.com/in/mounica-paladugu](https://www.linkedin.com/in/mounica-paladugu) | mounica-000.github.io

EDUCATION

Drexel University

Bachelor of Science in Computer Science, Minor in Mathematics

Philadelphia, PA

Sep 2020 – Jun 2025

WORK EXPERIENCE

Software Engineer

Oct 2025 – Present

Develop For Good

Remote, CA

- Led **end-to-end development** of a food recovery platform (**React, Python, FastAPI, Vercel**) serving Ohio organizations, shipping from **zero to production** in 3 months.
- Designed a **headless CMS architecture** using **Airtable** as a dynamic data layer, eliminating client dependency on engineers for content updates and cutting change request turnaround from **days to minutes**.
- Drove post-launch growth to **25+ MAU**(Monthly Active Users) within 2 weeks by integrating **Google Analytics** to identify friction points and iterate on UX.

Backend Software Engineer Intern

Mar 2023 – Sep 2023

Rajant Corporation

Malvern, PA

- Identified **CockroachDB** as a latency bottleneck for massive time-series health data; researched, evaluated, and migrated to **InfluxDB**, reducing query latency **~6x (350–500ms → 50–90ms)** and improving system scalability and reliability.
- Designed a **RESTful API (Python/FastAPI, Docker)** with **10+ endpoints** supporting multi-resolution health data aggregations (daily, weekly, monthly, 6-month), ingesting **5 wearable health measurements** and enabling integration with hospital records and the internal AI platform.
- Automated a real-time **file sync pipeline in Python** that watched a local directory and propagated changes to **MinIO S3** cloud storage automatically, eliminating **5+ hours of manual work per week**.

Product Development Engineer Intern

Mar 2022 – Sep 2022

Pression LLC

Coatesville, PA

- Automated the test data pipeline in **Python** for the company's medical device, reducing file storage costs by **25%** and cutting data loss by **50%**.
- Led the medical device transition from an unstable v1 prototype to a **reliable manufacturing-ready v2**, integrating **embedded C firmware** with system UI software, achieved **zero field failures** across all test scenarios, and enabled the company to pursue **FDA clearance filing**.
- Refactored and debugged **2000+ lines of broken UI code** and authored comprehensive system documentation covering architecture, data flows, and component design to support long-term maintainability.

PROJECTS

GitHub Issue Root Cause Agent | *Python, FastAPI, ChromaDB, Streamlit, Groq API, sentence-transformers* Mar 2026

- Built an end-to-end **RAG (Retrieval-Augmented Generation) pipeline** that ingests GitHub issues, generates **vector embeddings** using **all-MiniLM-L6-v2**, and stores them in a **ChromaDB** vector store for persistent semantic search.
- Engineered a **semantic retrieval system** using **cosine similarity** to surface the top-3 most relevant past issues for any new bug report, achieving **89% retrieval precision** across paraphrased query variants in end-to-end evaluation.
- Integrated **Llama 3.3 70b** via **Groq API** with a structured prompt template grounded in retrieved context, returning actionable root cause analysis and fix suggestions; exposed as a **REST API** via **FastAPI** with a **Streamlit** demo UI.

Virtual Pet | *TypeScript, React, HTML/CSS, MySQL, Node.js, WebSockets*

Mar 2025 – May 2025

- Built a **full-stack real-time application** with **WebSocket**-based communication, normalized relational schema, and persistent session management.
- Deployed on a **Linux** (Linode) cloud server using **React, TypeScript, and MySQL**.

Bank System | *Java, JUnit, Git, CI/CD, Object Oriented Programming (OOP)*

Mar 2024 – May 2024

- Implemented an **OOP** bank account management system in **Java** supporting APR calculations, validated through **TDD** (unit, integration, and system tests).
- Set up **Git-based CI/CD pipelines** to automate builds, testing, and secure deployment workflows.

SKILLS

Programming Languages: Python, Java, C, JavaScript, TypeScript, SQL, Bash, HTML/CSS

Frameworks & Libraries: FastAPI, Node.js, React, Express, JUnit

Databases & Tools: PostgreSQL, MySQL, Git, Docker, Postman, Vim, Figma

Systems & Practices: Linux/Unix, CI/CD, Agile, DevOps, SDLC, Jira, AI